Published online: 2021-10-08

Efficacy of Norway Spruce Ointments and Bacterial and Fungal Alterations in the Treatment of Castration Wounds in Piglets

Authors

Désirée Prokop[1], Joachim Spergser[2], Werner Hagmüller[3], Alexander Tichy[4], Karin Zitterl-Eglseer[1]

Affiliations

1 Institute of Animal Nutrition and Functional Plant Compounds, Department for Farm Animals and Veterinary Public Health, University of Veterinary Medicine, Vienna, Austria
2 Institute of Microbiology, Department of Pathobiology, University of Veterinary Medicine, Vienna, Austria
3 Institute of Organic Farming and Farm Animal Biodiversity, HBLFA Raumberg-Gumpenstein, Thalheim/Wels, Austria
4 Bioinformatics and Biostatistics Platform, Department of Biomedical Sciences, University of Veterinary Medicine, Vienna, Austria

Key words

Picea abies, Pinaceae, wound healing, resin, fungi, pig

received February 27, 2021
accepted after revision September 15, 2021
published online

Bibliography

Planta Med 2021
DOI 10.1055/a-1646-2959
ISSN 0032-0943

Georg Thieme Verlag KG, Rüdigerstraße 14, 70469 Stuttgart, Germany

Correspondence

Ao. Univ.-Prof. Mag. Dr. Karin Zitterl-Eglseer
Department for Farm Animals and Veterinary Public Health, Institute of Animal Nutrition and Functional Plant Compounds, University of Veterinary Medicine
Veterinaerplatz 1, 1210 Vienna, Austria
Phone: + 43 12 50 77 31 05, Fax: + 43 12 50 77 31 90
karin.zitterl@vetmeduni.ac.at

Supplementary material is available under https://doi.org/10.1055/a-1646-2959

ABSTRACT

This study aimed to evaluate the efficacy of Norway spruce ointments on wound healing of castration wounds in piglets. This study included 95 male pigs randomly divided into five treatment groups: Norway spruce balm (Vulpuran), Norway spruce resin (Abilar), pork lard (ointment base of Vulpuran), no treatment (negative control) and antibiotic blue spray (Cyclo spray, positive control). Wound healing parameters (such as healing time, wound size, reddening of wound edges and surrounding, swelling, secretion and wound contamination), microbiological status and the haptoglobin level as an inflammation parameter were investigated. In the Norway spruce groups, some positive effects on wound healing parameters were found. In the first 6 days of treatment, Abilar or Vulpuran showed the smallest means of wound areas, and at the end of the study (day 15 + 17), the highest rates of completely closed wounds compared to the other groups. Vulpuran treatment led to significantly lower wound secretion ($p = 0.003$) and wound contamination ($p = 0.015$) than the untreated control did. Furthermore, the microbiological status was determined using MALDI-TOF-MS and partial 16S rRNA gene sequencing at different days of treatment. A comparison of the five treatment groups on day 3 revealed that Norway spruce led to the lowest rate of wounds colonised with fungi, mainly classified into genus Candida, (Abilar 77 %, Vulpuran 70 %) in comparison with blue spray (89 %), lard (100 %) and untreated control (100 %). Fungi could only be detected in one of the 13 samples treated with Vulpuran on day 8, which nearly reached significance ($p = 0.055$).

ABBREVIATIONS

DHAA	dehydroabietic acid
EP	European Pharmacopoeia
MALDI-TOF MS	matrix-assisted laser desorption/ionisation-time-of-flight mass spectrometry

Introduction

The interest in phytotherapeutic products for wound healing, especially of poorly healing, chronic wounds, burns and other skin diseases, has increased in human and veterinary medicine. Wound healing is a complex process that involves chemotaxis, phagocytosis, cell division, neovascularization, synthesis of new extracellular matrix and formation of scar tissue [1]. Healing of acute wounds is divided into four, partial overlapping, phases: haemostasis, inflammation, proliferation, and remodelling. Immediately after an injury, the coagulation cascade starts, and clot formation and platelet aggregation stop the blood loss. In the inflammatory phase, neutrophil granulocytes infiltrate the wound, and the phagocytosis of bacteria and other foreign particles starts. The proliferation phase is characterized by the migration of fibroblasts, regeneration of the extracellular matrix, and tissue repair. Remodelling, the final phase of wound healing, is responsible for the development of new epithelium and formation of scar tissue [1, 2].

Numerous studies have shown various effects of plants and biological products on wound healing: One of the best-known substances is medicinal honey, which has antibacterial [3, 4] and antifungal [5] effects. Sterilized honeys achieved good results in therapy for burn wounds [6] and colonized or infected wounds that do not respond well to treatment [7]. Oryan and Zaker [8] treated a 3-cm incision of 40 White New Zealand rabbits by topical application of pure unheated honey. Treated lesions showed less oedema, less necrosis, better wound contraction, improved epithelialization, fewer polymorphonuclear and mononuclear cell infiltration and lower glykosaminoglykan and proteoglycan concentration levels [8].

Moreover, lavender essential oil has anti-inflammatory and analgesic properties [9, 10] as well as antimicrobial activity [11]. Lusby et al. [12] compared the treatment effects of Lavandula x allardii honey and essential oil and a standard therapeutic honey (Medihoney) for wound healing in rats. No significant difference was found in the wound contraction between the essential oil- or honey-treated wounds and the untreated wounds. Both honeys were shown to reduce the capillary volume in the wound site at day 12. In a veterinary study, Nada et al. [13] compared the efficacy of lavender oil with black seed oil, ostrich oil, and cod liver oil on surgically induced full-thickness skin wounds (2 cm width × 3 cm length) in the back region of dogs. Treatment with lavender oil showed reduction in wound dimension and good healing parameters compared with other oils, characterized by absence of inflammatory signs, exudation, and infection.

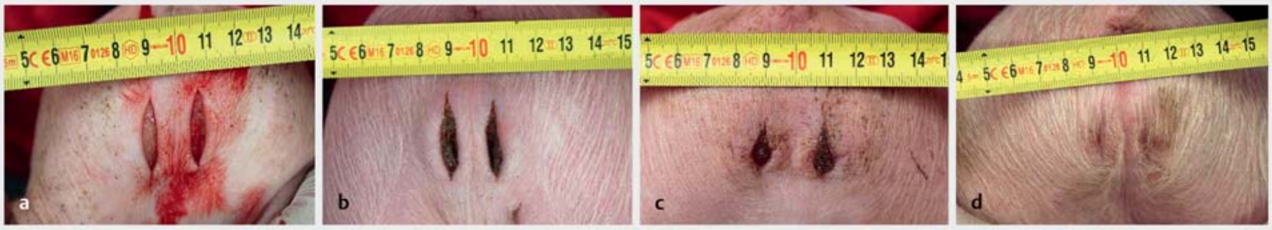
Spruce resins have been used for centuries in folk medicine for the treatment of skin injuries and are currently becoming the focus of interest in human medicine for the treatment of poorly healing chronic wounds (such as in pressure sores and diabetes) [14–16]. A traditionally produced ointment with Norway spruce resin was found to have bacteriostatic effects, primarily against Gram-positive bacteria including methicillin-resistant *Staphylococcus aureus* (MRSA), while Gram-negative bacteria showed less sensitivity to this ointment *in vitro*, with the exception of *Proteus vulgaris* [17, 18]. Changes in the bacterial cell wall and bacterial membrane and, as a result, a disturbance of the energy synthesis in the bacteria could be determined as the mechanisms of action [19]. Fungistatic effects of Norway spruce resin were also found *in vitro* [20], and especially against onychomycosis *in vivo* [21, 22].

Norway spruce balm is obtained from *Picea abies* (L.) H.Karst. (Pinaceae) by scraping the balm off the tree or by removing it from cavities. The balm contains diterpene resin acids, phenolic acids, lignans and essential oil [23], but the exact chemical composition of Norway spruce balm was not known until Goels et al. [24] described new aspects of the composition and effect of this phytochemical product on wound healing. They could isolate hydroxycinnamic acids (i.e., p-coumaric acid, caffeic acid and ferulic acid), the lignan pinoresinol, four hydroxylated derivatives of dehydroabietic acid (DHAA; i.e., 7α,15-dihydroxy-DHAA, 7β,15-dihydroxy-DHAA, 15-hydroxy-DHAA and 7-hydroxy-DHAA), DHAA itself and further diterpene acids without hydroxyl groups like abietic acid, neoabietic acid, pimaric acid, isopimaric acid, levopimaric acid, sandaracopimaric acid, and palustric acid. Furthermore, the compounds were tested for increased re-epithelialization of cell-free areas in a human adult keratinocyte monolayer. However, *in vitro* testing of skin preparations is a suboptimal approach, since the cellular test systems are water-based, but topical application requires lipophilic agents to permeate living skin. Simple *in vitro* models are not able to reproduce the complex settings of an *in vivo* model; the present study therefore focused on the treatment of castration wounds in male piglets. The skin of pigs can be considered a relevant model for humans because of the similar texture. Two finished medical devices were tested: Vulpuran (20 % Norway spruce balm in lard) and Abilar (10 % Norway spruce resin in paraffin) were compared to the antibiotic Cyclo spray (positive control), pork lard (excipient in Vulpuran) and control (without treatment) in terms of wound healing parameters. Wound area, day of complete wound closure, reddening of the wound edges and surrounding, wound swelling, secretion, contamination, the inflammation parameter haptoglobin, and certain parameters of differential blood count were evaluated. The microbiological status was determined using MALDI-TOF-MS and partial 16S rRNA gene sequencing. We hypothesized that the use of Norway spruce balm (Vulpuran) and Norway spruce resin (Abilar) leads to faster wound closure with less wound infection than the use of lard or non-treatment, revealing the spruce products to be equal to the antibiotic blue spray (Cyclo spray).

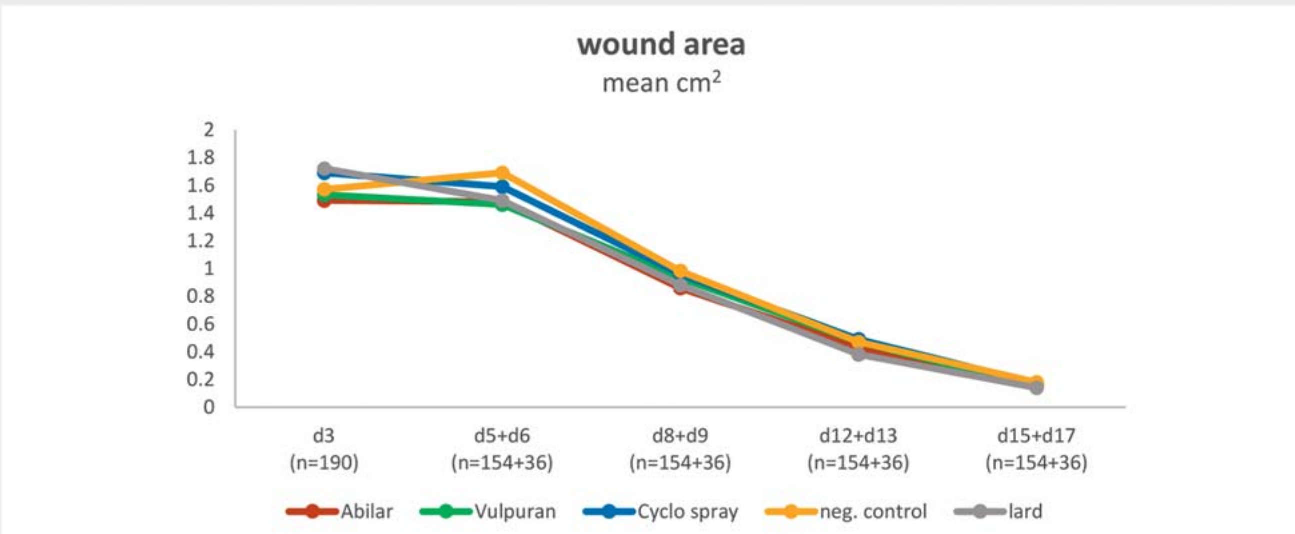
Results

The wound closure was monitored by photos which were taken throughout the whole healing process starting on day 1, the castration day. After the incision, the wounds averaged 2.2–2.8 cm in length, 0.2–0.6 cm in width and 0.5–1.3 cm^2 in wound area (see ▶ **Fig. 1**). Further evaluation was based on a total of 647 photos (each pig had 2 wounds, yielding 1294 wounds), which were analyzed for length, width, and wound area using the wound documentation programme WinDIA.

Because technical reasons made it impossible to inspect the wound in all replications on the same day, the results of day 5 and day 6 (d5+d6), d8+d9, d12+d13, as well as d15+d17, were considered as one data set including 95 pigs each. Beginning with d12, some wounds closed completely. Only the area of wounds not completely closed were measured from point day on. The timeline of the mean of the wound areas (cm^2) is depicted in ▶ **Fig. 2**. In the first 6 days of treatment Abilar (1.49 cm^2) and Vulpuran group (1.50 cm^2) showed the smallest mean of wound areas compared to the other groups (Cyclo spray 1.64 cm^2; neg. control 1.63 cm^2; lard 1.60 cm^2). However, the differences among the five treatment groups in terms of the size of the wound areas were not significant.

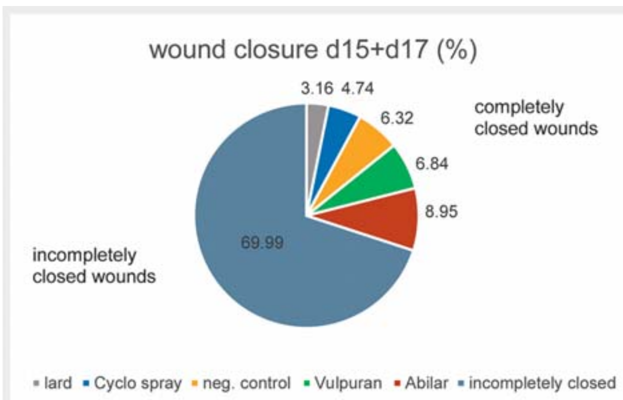


▶ **Fig. 1** Photos of wound on day 1 (castration day) (**a**), and wounds treated with Vulpuran on day 3 (**b**), day 8 (**c**), and day 15 (**d**).



▶ **Fig. 2** Timeline of wound areas (mean in cm²) in different groups (n = 190 wounds, y-axis = wound area in cm²).

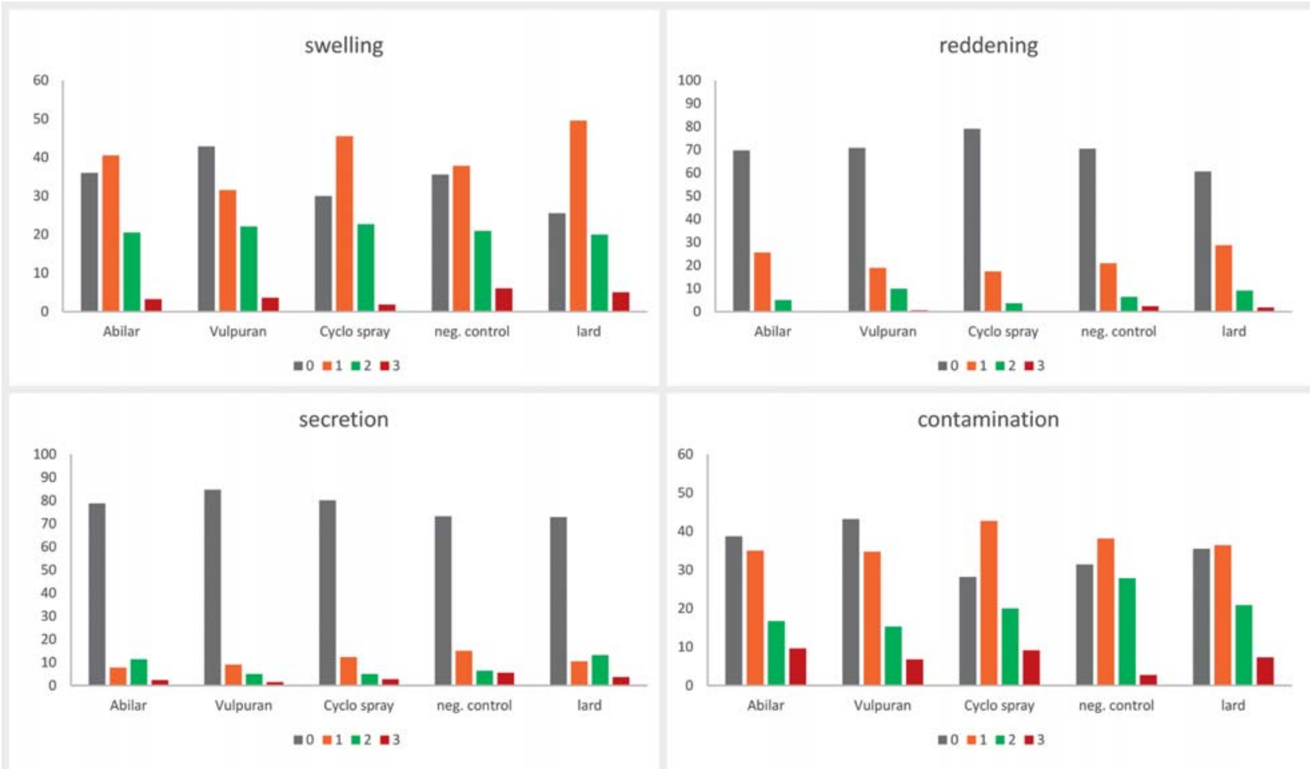
At the endpoint of the study (d15+d17), the highest rates of completely closed wounds were found in the Norway spruce groups Abilar (17 of 190 wounds; 8.95%) and Vulpuran (13 of 190 wounds; 6.84%). In the group receiving no treatment, 12 of 190 wounds (6.32%) and in the Cyclo spray group 9 of 190 wounds (4.74%) were completely closed, respectively. The lowest percentage of completely closed wounds was found in the lard group with 3.16% (6 of 190 wounds) (▶ **Fig. 3**).



▶ **Fig. 3** Wound closure (completely and incompletely closed wounds) in different groups at the end of the study (d15+d17) in percent, n = 190 wounds of 95 piglets.

The wound score was used to evaluate swelling of wound edges, reddening of the wound, secretion, and contamination with straw or other substances as absent, low, moderate, or high. The Vulpuran group showed the highest rates of inconspicuous wounds in terms of swelling of wound edges (95 of 220 wounds; 42.8%), wound secretion (188 of 220 wounds; 84.7%) and contamination (96 of 220 wounds; 43.2%). Only three wounds treated with Vulpuran showed high-grade secretion. No wound showed high-grade reddening with Abilar or blue spray, and in the Vulpuran group, one wound showed high-grade reddening. Most of the wounds without reddening could be found in the blue spray group (174 of 220 wounds; 79.1%) (▶ **Fig. 4**).

Statistical analyses revealed a significantly lower wound secretion ($p = 0.003$) and lower contamination ($p = 0.015$) in the Vulpuran group than in the untreated group. The Vulpuran group showed significantly less swelling ($p = 0.013$), less reddening ($p = 0.039$), and less secretion ($p = 0.001$) than did the lard group.



▶ **Fig. 4** Swelling, reddening, secretion and contamination of wounds (n = 220 wounds in each treatment group at all days except day 1; y-axis = number of wounds); 0 = not conspicuous, 1 = low irritation, 2 = moderate irritation, 3 = high irritation).

Wounds treated with blue spray showed significantly less reddening ($p = 0.000$) and less wound secretion ($p = 0.041$) than the wounds treated with lard, and significantly less reddening ($p = 0.023$) than the wounds treated with Vulpuran. The Vulpuran treated wounds showed significantly less contamination ($p = 0.002$) compared to blue spray. Wounds treated with Abilar showed significantly less reddening than the lard group ($p = 0.023$). No significant differences were found in all parameters, when comparing Abilar with Vulpuran.

Altogether, 1445 representative bacterial and 106 fungal colonies isolated from swab samples were investigated. The predominant phylum isolated was *Firmicutes* (n = 741, 51.3 % of isolates), followed by *Proteobacteria* (n = 402, 27.8 %), *Actinobacteria* (n = 221, 15.3 %), and *Fusobacteria* (n = 81, 5.6 %) (▶ **Table 1**). Bacterial isolates mainly belonged to orders *Lactobacillales* (*Firmicutes*, n = 298, 20.6 %), *Bacillales* (*Firmicutes*, n = 297, 20.5 %), *Enterobacterales* (*Proteobacteria*, n = 289, 20 %), *Actinomycetales* (*Actinobacteria*, n = 215, 14.9 %), and *Clostridiales* (*Firmicutes*, n = 146, 10.1 %). Most of the bacterial isolates at the family level belonged to *Staphylococcaceae* (*Firmicutes*, n = 238, 16.5 %), followed by *Corynebacteriaceae* (*Actinobacteria*, n = 177, 12.2 %), *Enterobacteriaceae* (*Proteobacteria*, n = 167, 11.5 %), *Streptococcaceae* (*Firmicutes*, n = 163, 11.3 %), *Aerococcaceae* (*Firmicutes*, n = 130, 9 %), and *Morganellaceae* (*Proteobacteria*, n = 108, 7.5 %). Most of the fungal isolates belonged to phylum *Ascomycota* (n = 104, 98.1 % of colonies) and order *Saccharomycetales* (n = 79, 74.5 %), followed by *Eurotiales* (n = 12, 11.3 %). More than half of the fungal colonies were classified into genus *Candida* (n = 71, 67 %).

On day 1, swabs were taken prior to incision for castration to assess the natural skin flora of the scrotum of the piglets (**Table 1S**, Supporting Information). Overall, on days 3 and 8, swab samples were taken from the wounds. On day 3, potentially pathogenic bacteria were more frequently found (**Table 2S**, Supporting Information), and on day 8 (**Table 3S**, Supporting Information), the flora shifted towards the skin flora of day 1. To define the effects of different treatments on the microbial composition of the wounds, the five treatment groups were statistically compared in terms of the evaluation parameters measured on days 3 and 8. For a clearer statistical evaluation, bacterial and fungal isolates were grouped according to taxonomic ranks (e.g., phylum or order affiliation). A comparison of the five treatment groups on day 3 revealed that the Vulpuran group showed the lowest rate of wounds colonized with fungi (70 %). A similar result was achieved by the Abilar group (77 %). By contrast, 88.9 % of the wounds in the blue spray group and 100 % of the wounds in the untreated and lard groups were colonized with fungi. The Vulpuran group also exhibited the lowest rate of samples from which *Candida famata* was isolated (60 %). The Vulpuran group also achieved the lowest results compared with the other four treatment groups, in the detection of fungal genera on day 8. Fungi could only be detected in one of the 13 samples, although this result did not reach significance ($p = 0.055$). On day 3, the five treatment groups showed significant differences in the occurrence of certain bacterial taxa

▶ **Table 1** Predominant bacterial and fungal taxa isolated from all piglets (bacteria, n = 267; fungi, n = 177).

Phylum (number of isolates)	Order (number of isolates)	Family (number of isolates)	Species predominantly isolated (number of isolates)
Firmicutes (741)	*Lactobacillales* (298)	*Streptococcaceae* (163)	*Streptococcus dysgalactiae* ssp. *equisimilis* (85)
			Streptococcus suis (68)
			Streptococcus spp.[a] (10)
		Aerococcaceae (130)	*Aerococcus viridans* (110)
			Facklamia tabacinasalis (20)
		Enterococcaceae (4)	*Enterococcus faecalis* (4)
		Lactobacillaceae (1)	*Lactobacillus casei* (1)
	Bacillales (297)	*Staphylococcaceae* (238)	*Staphylococcus hyicus* (82)
			Staphylococcus xylosus (56)
			Staphylococcus chromogenes (35)
			Staphylococcus equorum (32)
			Mammaliicoccus lentus (16)
			Staphylococcus spp.[b] (9)
			Jeotgalicoccus psychrophilus (6)
			Macrococcus caseolyticus (2)
		Bacillaceae (58)	*Bacillus licheniformis* (41)
			Lysinibacillus fusiformis (14)
			Bacillus pumilus (2)
			Bacillus pseudomycoides (1)
		Planococcaceae (1)	*Kurthia populi* (1)
	Clostridiales (146)	*Peptostreptococcaceae* (81)	*Peptostreptococcus canis* (81)
		Clostridiaceae (65)	*Clostridium perfringens* (65)
Proteobacteria (402)	*Enterobacteriales* (289)	*Enterobacteriaceae* (167)	*Escherichia coli* (155)
			Klebsiella oxytoca (4)
			Salmonella enterica (2)
			Escherichia fergusonii (1)
			Enterobacter ludwigii (1)
			Enterobacter absuriae (1)
			Klebsiella pneumoniae (1)
			Serratia marcescens (1)
			Citrobacter braakii (1)
		Morganellaceae (108)	*Proteus mirabilis* (91)
			Providencia rettgeri (7)
			Proteus hauseri (6)
			Proteus vulgaris (3)
			Morganella morganii (1)
		Erwiniaceae (14)	*Pantoea agglomerans* (13)
			Pantoea ananatis (1)

continued

▸ **Table 1** Predominant bacterial and fungal taxa isolated from all piglets (bacteria, n = 267; fungi, n = 177).

Phylum (number of isolates)	Order (number of isolates)	Family (number of isolates)	Species predominantly isolated (number of isolates)
	Pseudomonadales (93)	*Moraxellaceae* (77)	*Psychrobacter pasteurii* (44)
			Acinetobacter gandensis (11)
			Acinetobacter johnsonii (8)
			Acinetobacter towneri (7)
			Acinetobacter spp.[c] (4)
			Moraxella pluranimalium (2)
			Psychrobacter pulmonis (1)
		Pseudomonadaceae (16)	*Pseudomonas stutzeri* (14)
			Pseudomonas spp.[d] (2)
	Pasteurellales (12)	*Pasteurellaceae* (12)	*Mannheimia haemolytica* (4)
			Pasteurella aerogenes (3)
			Actinobacillus seminis (3)
			Mannheimia varigena (2)
	Lysobacterales (3)	*Lysobacteraceae* (3)	*Stenotrophomonas maltophilia* (3)
	Burkholderiales (2)	*Alcaligenaceae* (2)	*Alcaligenes faecalis* (2)
	Rhizobiales (2)	*Rhizobiaceae* (2)	*Rhizobium* spp.[e] (2)
	Incertae sedis (1)	*Incertae sedis* (1)	*Wohlfahrtiimonas chitiniclastica* (1)
Actinobacteria (221)	*Actinomycetales* (215)	*Corynebacteriaceae* (177)	*Corynebacterium xerosis* (82)
			Corynebacterium stationis (43)
			Corynebacterium casei (18)
			Corynebacterium amycolatum (10)
			Corynebacterium glutamicum (10)
			Corynebacterium humireducens (7)
			Corynebacterium spp.[f] (7)
		Micrococcaceae (19)	*Micrococcus lylae* (9)
			Rothia nasimurium (7)
			Arthrobacter oryzae (3)
		Actinomycetaceae (10)	*Trueperella pyogenes* (10)
		Dermabacteraceae (8)	*Brachybacterium muris* (8)
		Dietziaceae (1)	*Dietzia aerolata* (1)
	Coriobacteriales (6)	*Coriobacteriaceae* (6)	*Atopobium fossor* (6)
Fusobacteria (81)	*Fusobacteriales* (81)	*Fusobacteriaceae* (81)	*Fusobacterium necrophorum* (81)

continued

▶ **Table 1** Predominant bacterial and fungal taxa isolated from all piglets (bacteria, n = 267; fungi, n = 177).

Phylum (number of isolates)	**Order (number of isolates)**	**Family (number of isolates)**	**Species predominantly isolated (number of isolates)**
Ascomycota (fungal) (104)	*Saccharomycetales* (79)	*Incertae sedis* (71)	*Candida famata* (71)
		Dipodascaceae (8)	*Geotrichum candidum* (8)
	Eurotiales (12)	*Trichocomaceae* (12)	*Aspergillus niger* (5)
			Aspergillus flavus (4)
			Penicillium chrysogenum (2)
			Aspergillus fumigatus (1)
	Hypocreales (8)	*Nectriaceae* (6)	*Fusarium solani* (6)
		Cordycipitaceae (2)	*Akanthomyces muscarius* (1)
			Beauveria pseudobassiana (1)
	Microascales (4)	*Microascaceae* (4)	*Scopulariopsis brevicaulis* (4)
	Pleosporales (1)	*Pleosporaceae* (1)	*Alternaria alternata* (1)
Zygomycota (fungal) (2)	*Mucorales* (2)	*Mucoraceae* (2)	*Rhizopus microsporus* (1)
			Mucor racemosus (1)

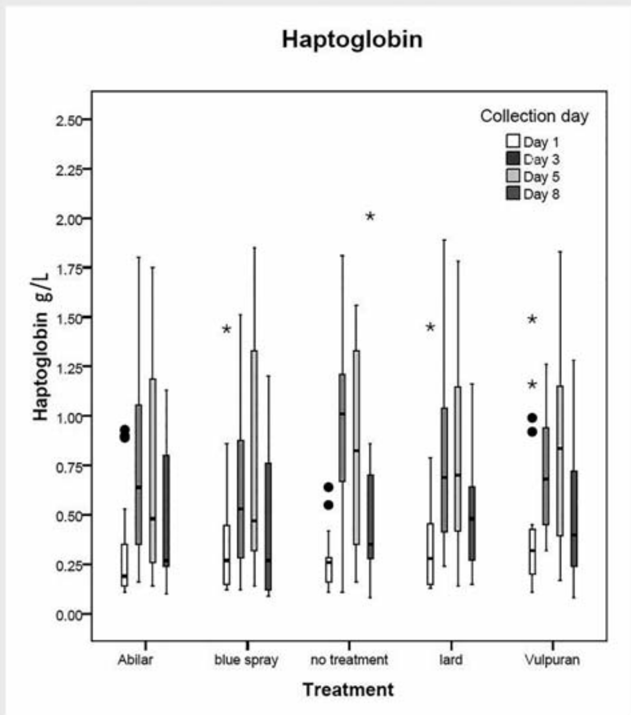
[a] *Streptococcus* (*Sc.*) spp. including *Sc. oralis* (1), *Sc. canis* (1), *Sc. henryi* (1), *Sc. orisratti* (1), *Sc. alactolyticus* (2), *Sc. pluranimalium* (2), *Sc. gallolyticus* (1) and *Sc. mitis* (1); [b] *Staphylococcus* (*St.*) spp. including *St. hominis* (3), *St. haemolyticus* (3), *St. cohnii* (1), *St. microti* (1) and *Mammaliicoccus sciuri* (1); [c] *Acinetobacter* (*A.*) spp. including *A. guillouiae* (1), *A. lwoffii* (2) and *A. baumanii* (1); [d] *Pseudomonas* (*P.*) spp. including *P. koreensis* (1) and *P. chlororaphis* (1); [e] *Rhizobium* (*R.*) spp. including *R. radiobacter* (1) and *R. pusense* (1); [f] *Corynebacterium* (*C.*) spp. including *C. faecale* (2), *C. ammoniagenes* (3), *C. confusum* (1) and *C. efficiens* (1)

and microbial compositions. In the blue spray group, *Staphylococcaceae* (p = 0.04), *Staphylococcus (St.) hyicus* (p = 0.045), *Clostridium perfringens + Fusobacterium + Peptostreptococcus* (anaerobes) (p = 0.033), and *Fusobacterium + Peptostreptococcus* (p = 0.033) were significantly less detected. On day 8, a numerical difference was noted in the Vulpuran group, in which no sample was positive for *Pseudomonadales*, and this group showed the lowest percentage of samples from which *Proteobacteria* was isolated (42.1 %), compared with the untreated group (47.4 %), lard group (63.2 %), and blue spray group (57.9 %). No significant differences were found in the occurrence of other bacterial taxa among the groups.

A total of 332 blood samples were analyzed for haptoglobin (▶ **Fig. 5**). Haptoglobin is an acute phase protein which indicates inflammation signs of the wounds. The acute phase proteins react early to systemic diseases, the most illnesses with inflammatory reactions lead to a rise of acute phase protein [25]. The haptoglobin concentrations depend on the age of the pigs; age-independent cut-off values for non-specific inflammation of 0.85 mg/mL could be taken as limit value [25, 26]. In our study, Laboklin (Linz, Austria) set a limit of haptoglobin values of 0.68 g/L. As baseline values, 12.6 % of samples with haptoglobin values > 0.68 g/L could be determined on day 1 (highest value with 1.49 g/L), and 87.4 % of the samples (n = 95) had haptoglobin concentrations below the baseline. On day 3, 46.3 % of the samples (n = 95) had haptoglobin values below and 53.7 % had values higher than the baseline (highest value of 1.89 g/L). On day 5, 49.4 % of the samples (n = 77) had haptoglobin values < 0.68 g/L and 50.6 % had values higher than the baseline (highest value of 1.85 g/L). On day 8, 70.8 % of the samples (n = 65) had haptoglobin values below the baseline, and 29.2 % had values higher than the baseline (highest value with 2.01 g/L). No significant differences in haptoglobin levels were found between the treatment groups on treatment days (▶ **Fig. 5**).

By contrast, significant differences were found within each treatment group in a daily comparison of haptoglobin levels, with the lowest values on day 1 before castration (Vulpuran, day 1/day 3, p = 0.026, day 1/day 5, p = 0.018; Abilar, day 1/day 3, p = 0.004, day 1/day 5, p = 0.017; blue spray, day 1/day 3, p = 0.043, day 1/day 5, p = 0.027; untreated control, day 3/day 8, p = 0.023; lard, day 1/day 3, p = 0.003, day 1/day 5, p = 0.011, day 3/day 8, p = 0.039). Thus, greater differences in the haptoglobin values could be found between the individual collection days, especially between day 1 and day 3 and between day 1 and day 5, than between each treatment group.

Neither the treatment groups nor the individual collection days showed significant difference in leukocyte levels. As regarding platelet levels, no difference was found among treatment groups, but a significant difference was noted in the collection days (p = 0.017) in all groups. Platelets were the highest at day 5, measured across all groups, with significant differences at day 1/day 5 (p = 0.011) and day 3/day 5 (p = 0.003). Segment-nucleated granulocytes showed no changes on the individual collection days, but a significant difference (p = 0.032) was observed between the blue spray group with lower values and lard group over all collection days. In addition, lymphocyte values showed no changes over the study period, but significant differences were found between the lard group (fewer lymphocytes were measured over all collection days) and blue spray group (p = 0.039),



▸ **Fig. 5** Haptoglobin levels (comparison within the treatment groups at days 1, 3, 5, 8). Variables for the box plots in SPSS: boxes represent values in the range from the 1–3 quartile; edges of the boxes = 1. quartile and 3. quartile; whiskers = range is defined by difference between the largest and smallest outliers; dots = outliers = values with a distance to 1. and 3. quartile of < 1.5 × interquartile range (IQR); asterisks = extreme values with a distance to 1. and 3. quartile of > 1.5 × IQR; line within the box = median

untreated control (p = 0.012), and Vulpuran group (p = 0.002). The levels of monocytes were the highest on day 8, and significant differences were found between day 1 and day 5 (p = 0.018), day 1 and d 8 (p = 0.000), day 3 and day 8 (p = 0.000), and day 5 and day 8 (p = 0.002), as well as between the Abilar group and Vulpuran group (p = 0.019), blue spray group, and Vulpuran group (p = 0.050), lard group and Vulpuran group (p = 0.001) and untreated control group and lard group (p = 0.007) in all collection days.

Discussion

The use of phytopharmaceuticals for wound healing disorders is becoming a general focus of interest because of the antibiotic resistance problem, not only in livestock, but in small animals and humans. In addition to existing *in vitro* studies on the use of Norway spruce resin for wound healing and its antimicrobial and antifungal effect [18, 20, 24, 27], the present study aimed to demonstrate the effect of Norway spruce balm ointment *in vivo* on wound healing in piglets to establish its use in practice. Sipponen et al. [28] described cases of three patients (with pressure ulcer on the heel, pressure ulcer in the sacral area and infected amputation stump) whose wounds did not heal over a long period of conventional wound treatment and antibiotics. By using Lappish spruce resin salve, all wounds closed completely at 6, 7 and 2 months, respectively. In a prospective, randomized and controlled multicenter trial from 2008, Sipponen et al. [14] compared the effect of resin salve with a control group treated with sodium carboxymethylcellulose hydrocolloid polymer in 37 patients with grade 2–4 pressure ulcers. All ulcers healed in 12 of 13 patients in the resin group, but only in 4 of 9 patients in the control group.

Vulpuran contains 20 % Norway spruce balm in pork lard, and Abilar, 10 % Norway spruce resin in a salve base (petrolatum, paraffinum liquidum, alcohol denat., cera microcristallina, sorbitan oleate, cera alba, hydrogenated castor oil, and stearic acid). Both are registered as medical devices class II b, which are meant to form a physical barrier against bacteria and yeasts. The active ingredient of Cyclo spray is 2.45 % w/w chlortetracycline HCl (excipients: patent blue V (E 131), butane 100, colloidal anhydrous silica, isopropyl alcohol, sorbitan trioleate). Chlortetracycline exerts its bacteriostatic action by inhibiting the protein synthesis of the bacterial cell.

In our study, no significant differences were found in the wound size between treatment groups on each collection day, but more wounds were completely closed with Vulpuran or Abilar ointment than with antibiotic spray, pure lard, and no treatment at the end of the study (d15+d17). Thus, we can confirm the results of the above-mentioned studies that Norway spruce resin or Norway spruce balm has positive effects on wound healing until complete wound closure. Comparing the results of Norway spruce products with those of Cyclo spray, we assume that acceleration of wound healing is not only due to the antimicrobial effect of Vulpuran and Abilar. In the assessment of wound healing parameters, a significantly milder wound secretion and contamination could be demonstrated in the Vulpuran group than in the untreated control group. Similarly, there was less frequent occurrence of wound edge swelling and occasional severe reddening of the wounds. This finding supports the adoption of the positive effect of Vulpuran on inflammatory and healing processes of the wounds, and partially confirms our hypothesis.

In an *in vitro* study, Rautio et al. [20] explored the antifungal effectiveness of resin from the Norway spruce through agar plate diffusion tests and electron microscopy. Resin salves were used in different concentrations (i.e., 10 %, 20 %, 30 % and 40 %) of purified resin mixed with animal fat or with a formulated modern salve base (i.e., sorbitan oleate 17 %, hydroxyethyl cellulose 2 % and water 81 %) and tested against yeasts, dermatophytes, and opportunistic fungi. Results indicated that resin possesses antifungal properties against dermatophytes including *Trichophyton* species but not against yeasts of genus *Candida* or opportunistic fungi. Electron microscopy provided evidence that resin caused damage to hyphae and cell wall structures. Furthermore, the antifungal activity of resin against *Trichophyton* species was dependent on its concentration.

Sipponen and Laitinen [27] examined the antimicrobial properties of Norway spruce resin against bacteria and yeasts using the European Pharmacopoeia (EP) challenge test. They found a microbicidal effect against *Candida albicans* and certain bacteria. However, they were unable to explain the discrepancies between results obtained by agar plate diffusion and EP challenge test, but they assumed that water solubility and diffusion could be the cause. Moreover, structural and biological differences between yeasts and dermatophytes may explain the varying effectiveness of resin [20]. Altogether, dermatophytes were shown to be susceptible to the antifungal effects of resin, indicating that a higher concentration of Norway spruce resin salves can be used to treat skin and nail infections caused by dermatophytes [20].

In an *in vivo* study [21], resin from the Norway spruce was used for topical treatment of 37 patients with onychomycosis, resulting in complete repair in three and partial healing in 11 patients; thus, in 14 patients, the treatment was considered clinically effective. At the beginning of the study, 20 of 37 patients had a positive mycological culture; at the end of the study, 13 of these patients had a negative culture, and in one case, the culture was initially negative but subsequently turned positive.

In our *in vivo* study, the Vulpuran and Abilar groups had the lowest percentage of wounds colonized with fungi (~ 70%) on day 3, whereas in the untreated group and lard group, all wounds (100%) were colonized with fungi. Interestingly, the Vulpuran group also contained the lowest percentage of samples positive for *Candida famata*. Moreover, on day 8, fungi were only detected in 1 of 13 samples in the Vulpuran group, supporting previous report on the antifungal properties of Norway spruce resin.

In the EP challenge test of Sipponen and Laitinen [27], Norway spruce resin was mixed with a biologically inert salve in different concentrations; a salve medium without resin was used as control. The following microbes were examined: *S. aureus*, MRSA, *Escherichia coli, Pseudomonas aeruginosa, Bacillus subtilis,* and *Candida albicans*. Pure resin (dissolved in ethanol) had antimicrobial activity against all tested microbial species within 1 day. No differences were observed between the 0.5% resin salve medium and the resin free salve medium, which apparently had no antimicrobial properties. With a minimum concentration of 10% of the resin to create a significant effect, increasing concentrations of the resin in the salve medium reduced the number of Gram-positive and Gram-negative bacteria. In an *in vitro* study [18], Norway spruce resin showed antimicrobial effects against Gram-positive bacteria, including *St. aureus* and MRSA in the agar diffusion test. No antimicrobial activities against Gram-negative bacteria were observed, except for *Proteus vulgaris*. Moreover, in the EP challenge test, Gram-positive bacteria were more rapidly killed, and a lower concentration of resin was required [27]. Using transmission and scanning electron microscopy, Sipponen et al. [19] studied the effects of resin on *S. aureus* and thereby demonstrated that resin may cause changes in the cell wall and cell membrane, affecting cell viability.

In our *in vivo* study, we could not detect any significant differences in the bacterial composition between the individual treatment groups. Only on day 3, *Staphylococcaceae, St. hyicus, Clostridium perfringens, Fusobacterium* and *Peptostreptococcus* were significantly less common in the blue spray group. However, these potentially pathogenic bacterial taxa were not frequently present in the untreated control group or in groups treated with spruce resin products.

In the evaluation of differential blood count and haptoglobin, we could not identify any significant differences between the five treatment groups. However, some differences were observed between collection days in one group, but to a normal extent. We observed some significant differences in the haptoglobin values between day 1 and day 3 as well as between day 1 and day 5, with mostly lower values on a day before castration. Thus, the haptoglobin value was probably influenced by the castration in some piglets, but not influenced by the allocated therapy.

Wounds treated with pure lard showed worse results in some parameters than did wounds with other treatments or untreated wounds, although lard is the ointment base of Vulpuran. The lard group had the fewest completely closed wounds at the end of the study (3.16%). Significant differences in the following parameters were found between the Vulpuran group and the lard group: redness of the wounds ($p = 0.039$), swelling of wound edges ($p = 0.013$) and secretion ($p = 0.001$). The results suggest that ointment base alone does not lead to faster wound healing or reduction of inflammatory parameters; thus, the Norway spruce balm is the effective ingredient that influences wound healing.

Furthermore, it would be of interest to study which active substances of the Norway spruce resin are responsible for the bacteriostatic, fungistatic, anti-inflammatory and in general wound healing properties. On rough average, 10% p-coumaric acid, 60% resins acids and 30% lignans were found in Norway spruce resin by means of GLC after silylation [20, 23, 27]. Sipponen et al. [19] observed diminution of cell viability in *St. aureus* not only for the resin, but in particular for abietic acid. Studying re-epithelialization of cell-free areas in a human adult keratinocyte monolayer Goels et al. [24] found a contribution of lignans and diterpene resin acids to increased migration and proliferation of keratinocytes upon exposure to spruce balm. However, re-epithelialization is just a small cut-out within the complex process of wound healing. For definition of active marker substances in Norway spruce products further studies concerning bioassay-guided isolation of the compounds as well as analytical methods for their accurate and validated quantification are required.

The results of this study showed Norway spruce balm ointment equal to conventional therapy with positive effects on selected wound healing parameters and less colonization of wounds with fungi.

Materials and Methods

Experimental setup

The experiment was carried out with organically reared piglets from the 'Institute of Organic Farming and Farm Animal Biodiversity' at the Thalheim/Wels branch of the HBLFA Raumberg-Gumpenstein, Austria, from April 2019 to August 2019. A total of 95 piglets, from crossbreeds of Large White and Landrace and Pietrain as final stage boar, were included in the study. The pigs were kept on straw in a farrowing unit, in a group size of 10 littermates (5 female, 5 male) and a mother sow. A heated creep area and an outdoor run were available in each pen. From the 3rd week of life, the piglets were fed with organic creep feed, and the sows were fed ad libitum with suckling feed (complete feed, Fixkraft). Sow and piglets had a mother–child drinker in the outdoor run, which was accessible ad libitum from the first day.

Only male animals were used for the study. The pigs were neutered at approximately age 3 weeks under general anaesthesia with 15 mg/kg ketamine (Ketasol, Richter Pharma AG), 0.2 mg/kg butorphanol (Butomidor, Richter Pharma AG) and 2 mg/kg azaperone (Stresnil, Elanco). Two incisions of approximately the same size were made on the scrotum. After castration, 0.4 mg/kg of meloxidolor (Metacam, Ceva Tiergesundheit GmbH) was applied to each animal for pain treatment.

On the castration day (day 1), the pigs were randomly divided into the five treatment groups: (1) Norway spruce balm ointment (Vulpuran, Lupuca Pharma GmbH), (2) Norway spruce resin ointment (Abilar, 10% resin salve, Repolar GmbH), (3) pork lard (ointment base of Vulpuran; placebo), (4) no treatment (negative control), and (5) antibiotic blue spray (Cyclo spray, Dechra Veterinary Products GmbH; positive control). A total of 95 piglets were included in the study, and a total of four series took place with the following subgroups: (1) Norway spruce balm ointment (n = 19, four series with 3, 6, 6 and 4 piglets), (2) Norway spruce resin ointment (n = 19, four series with 4, 6, 5 and 4 piglets), (3) pork lard (n = 19, four series with 4, 6, 5 and 4 piglets), (4) no treatment (n = 19, four series with 3, 6, 5 and 5 piglets), and (5) antibiotic spray (n = 19, four series with 4, 6, 5 and 4 piglets).

On days 1 (castration day), 2, 3, 4, 5, 8 and 9, 0.5 mL of each product was applied topically on each incision with a single-use syringe without a needle. In the antibiotic blue spray group (Cyclo spray), one push from the spray can was applied per incision. The wounds of the piglets were left untreated from day 10–17.

The study was approved by the institutional ethics and animal welfare committee and the national authority according to §§ 26 ff. of Animal Experiments Act, Tierversuchsgesetz 2012 – TVG 2012. Date of approval: February 12th, 2019, approval number: BMBWF GZ 66.019/0005-V/3b/2019.

Data collection

Photo measurement

Wounds of 77 piglets were photographed on days 1, 3, 5, 8, 10, 12, and 15. Photos of the remaining 18 piglets were taken on days 1, 3, 6, 9, 13 and 17. The photos were taken with a Canon EOS 2000D with an EF-S 18.55 mm f/3.5–5.6 IS II lens. A scale was used for orientation. Within this period, 647 photos were taken and then analyzed using a digital wound documentation programme (WinDIA). Both castration wounds were measured and evaluated independently by sizing the length (longest point), width (widest point), and wound area. With the help of WinDIA, the wound area could be rounded and measured exactly. For evaluation data of 77 piglets at day 5 and 18 piglets at day 6 (d5+d6), as well as at d8+d9, d12+d13 and d15+d17 were combined.

Wound score

Further evaluation of the wounds was based on the following parameters: reddening of wound edges and surrounding, swelling of the wound, secretion, and contamination with straw or other substances. All parameters were scored as 0 (not conspicuous), 1 (low irritation), 2 (moderate irritation), or 3 (high irritation).

Bacteriology

For bacteriological and mycological examinations, swab samples were taken from the intact scrotum (day 1) and castration wound (days 3 and 8) of 18 piglets (days 1 and 8) and 77 piglets (day 1, 3 and 8), respectively. In total, 267 samples were collected using sterile cotton swabs that were immediately placed into Amies transport medium (Transwab M40 compliant, Amies medium, sterile, Medical Wire & Equipment). The samples were stored and refrigerated (2–8 °C) until microbiological examination at the diagnostic laboratory of the Institute of Microbiology at the University of Veterinary Medicine Vienna was performed 3–5 h after sampling.

For microbiological examination, swabs were streaked onto the following agar plates to obtain isolated microbial colonies: two Columbia agar with 5% sheep blood (BBL, BD Diagnostics) incubated in ambient air or in an anaerobic jar with gas packs (BD Diagnostics) at 37 °C for 48 h, CN (Colistin-Nalidixic Acid) agar with 5% sheep blood (BBL) (medium for the selective isolation of Gram-positive bacteria) and MacConkey II agar (BBL) (medium for the selective isolation of Gram-negative bacteria), both incubated in ambient air at 37 °C for 48 h.

In addition, yeast and moulds were isolated from swab samples taken from 18 piglets on day 1 and day 8 and from 47 piglets on days 1, 3, and 8 using Sabouraud Dextrose agar with Emmons modification (BBL™) incubated aerobically at 28 °C for up to 14 days.

Bacterial isolates were first examined for colony morphological characteristics, and representatives of colony morphotypes were then identified using matrix-assisted laser desorption/ionization-time-of-flight mass spectrometry (MALDI-TOF MS) (Bruker Daltonics). Briefly, single bacterial colonies representing a distinct morphotype were smeared onto a 96-target polished steel plate using a toothpick and overlaid with 1 µL of 70% formic acid. After air-drying, spots were overlaid with 1 µL of α-cyano-4-hydroxycinnamic acid matrix solution (10 mg/mL in 50% acetonitrile and 2.5% trifluoroacetic acid), and mass spectra were generated using a microflex LT Biotyper operating system (Bruker Daltonics). Data were analyzed in automatic mode using Bruker FlexControl 3.4 software and MBT Compass Explorer 4.1 with an integrated taxonomy library containing 7854 reference spectra (Bruker Daltonics). For calibration, and as a quality control, a bacterial test standard was used in each run. The degree of spectral concordance was expressed as a logarithmic identification score, which was interpreted according to the manufacturer's instructions with score values ≥ 2.00 considered acceptable for the identification at the species level. For isolates with score values < 2.00, main spectrum profiles were created as described previously [29], and a study-related database was constructed, which was used in addition to the integrated taxonomy library of the Compass Explorer 4.1. Bacterial isolates with unique mass spectra or representatives of bacterial isolates unidentifiable at the species level using MALDI-TOF MS (< 2.00 using the taxonomy library of Compass Explorer 4.1) and expressing similar mass spectrometric profiles were finally classified by partial 16S rRNA gene sequencing.

Using UltraClean Microbial DNA Isolation Kit (Mo Bio Laboratories), DNA from single colonies were extracted and the 16S rRNA gene was amplified, employing primers 27 f and 1492 r and thermal cycling conditions as described by Lane [30]. PCR products were then purified using UltraClean PCR Clean-Up Kit (Mo Bio Laboratories) and sequenced in both directions at LGC Genomics. Overlapping regions of resultant sequences were aligned, and

▸ **Table 2** Summary of experimental groups, group size, and treatments at the different days.

	day	Vulpuran (pigs = n)	Abilar (pigs = n)	blue spray (pigs = n)	lard (pigs = n)	no treatment (pigs = n)	total (pigs = n)
wound assessment	1	19	19	19	19	19	95
	3	19	19	19	19	19	95
	5	16	15	15	15	16	77
	6	3	4	4	4	3	18
	8	16	15	15	15	16	77
	9	3	4	4	4	3	18
	10	16	15	15	15	16	77
	12	16	15	15	15	16	77
	13	3	4	4	4	3	18
	15	16	15	15	15	16	77
	17	3	4	4	4	3	18
bacteriology	1	19	19	19	19	19	95
	3	16	15	15	15	16	77
	8	19	19	19	19	19	95
yeast and moulds	1	13	13	13	13	13	65
	3	10	9	9	9	10	47
	8	13	13	13	13	13	65
haptoglobin	1	19	19	19	19	19	95
	3	19	19	19	19	19	95
	5	16	15	15	15	16	77
	8	13	13	13	13	13	65
blood count	1	9	10	10	10	9	48
	3	9	10	10	10	9	48
	5	6	6	6	6	6	30
	8	3	4	4	4	3	18

nearly complete 16S rRNA gene sequences were subjected to similarity search against the EzBioCloud database (http://www.ezbiocloud.net/identify) [31]. Sequence similarity values of ≥ 98.7 % [32] and ≥ 95 % [33] were used as cut-off values for species and genus affiliation.

Mould and yeast isolates were identified by micro- and macroscopic morphological examination and by sequencing of the internal transcribed spacer region (ITS1–5.8S-ITS2) of morphological representatives. For classification, obtained sequences were compared with entries in the GenBank nucleotide database using the BLASTN algorithm.

Haptoglobin

In 18 piglets on days 1, 3, and 8; in 30 piglets on days 1, 3, and 5; and in 47 piglets on days 1, 3, 5, and 8, 2 mL of blood was taken from the vena cava cranialis (VACUETTE tubes 2 mL CAT Serum Clot Activator 13 × 75). The blood samples were refrigerated and brought to Laboklin usually 2 h after sampling.

Blood count

In 18 piglets on days 1, 3, and 8; and in 30 piglets on days 1, 3, and 5, 1 mL of EDTA, blood was taken from the vena cava cranialis (VACUETTE tubes 3 mL K3EDTA 13 × 75). Blood samples were refrigerated and brought to Laboklin usually 2 h after sampling for determination of the differential blood count. The most important parameters in this study (i.e., leukocytes, platelets, segment-nucleated granulocytes, lymphocytes, and monocytes) were statistically evaluated.

A summary of experimental groups, group size and treatments at the different days is provided in ▸ **Table 2**.

Statistics

The analyses were performed using the IBM SPSS v24 programme (IBM Corp.). The different treatment groups were compared concerning the various scores of the wound parameters using the Mann-Whitney Test. General linear models were used to analyze the differences in blood parameters among treatment groups, and the wound area was included as a covariate in addition to the factor treatment, followed by post hoc tests using Sidak's alpha correction procedure. The assumption of normal distribution was tested using the Kolmogorov–Smirnov test. The Chi^2 test was used to analyse differences in the frequency distribution as regards the number of healed wounds among the groups. A p-value < 5 % ($p < 0.05$) was assumed to be significant for all statistical analyses.

Supporting Information

Tables 1S–3S show the predominant bacterial and fungal taxa isolated from the piglets before incision and after treatment of the wounds on days 3 and 8. **Table 1S**: Predominant bacterial and fungal taxa isolated from the scrotum of the piglets (bacteria, n = 95 pigs; fungi, n = 65 pigs) on day 1 (before castration). **Table 2S**: Predominant bacterial and fungal taxa isolated from the castration wounds of the piglets (bacteria, n = 77 pigs; fungi, n = 47 pigs) on day 3. **Table 3S**: Predominant bacterial and fungal taxa isolated from the castration wounds of the piglets (bacteria, n = 96 pigs; fungi, n = 65 pigs) on day 8.

Contributors' Statement

Data collection: D. Prokop, W. Hagmüller, J. Spergser, A. Tichy; design of the study: K. Zitterl-Eglseer, D. Prokop, J. Spergser, W. Hagmüller, A. Tichy; statistical analysis: A. Tichy; analysis and interpretation of the data: D. Prokop, J. Spergser, K. Zitterl-Eglseer, W. Hagmüller, A. Tichy; drafting the manuscript: D. Prokop, K. Zitterl-Eglseer, J. Spergser; critical revision of the manuscript: K. Zitterl-Eglseer, J. Spergser, W. Hagmüller.

Acknowledgements

This work was supported by the employees of the Thalheim/Wels branch of the HBLFA Raumberg-Gumpenstein (Austria) by helping with the castration and care of the pigs and the sampling. We would like to thank the Lupuca company, which provided the Vulpuran ointment free of charge for this project. The authors would like to thank Enago (www.enago.com) for the English language review.

Conflict of Interest

The authors declare that they have no conflict of interest.

References

[1] Enoch S, Leaper DJ. Basic science of wound healing. Surgery 2008; 26: 31–37

[2] Velnar T, Bailey T, Smrkolj V. The wound healing process: an overview of the cellular and molecular mechanisms. J Int Med Res 2009; 37: 1528–1542

[3] Cavanagh D, Beazley J, Ostapowicz F. Radical operation for carcinoma of the vulva. A new approach to wound healing. J Obstet Gynaecol Br Commonw 1970; 77: 1037–1040

[4] Mandal MD, Mandal S. Honey: Its medicinal property and antibacterial activity. Asian Pac J Trop Biomed 2011; 1: 154–160

[5] Irish J, Carter DA, Shokohi T, Blair SE. Honey has an antifungal effect against Candida species. Med Mycol 2006; 44: 289–291

[6] Moore OA, Smith LA, Campbell F, Seers K, McQuay HJ, Moore RA. Systematic review of the use of honey as a wound dressing. BMC Complement Altern Med 2001; 1: 2

[7] Dunford C, Cooper R, Molan P, White R. The use of honey in wound management. Nurs Stand 2000; 15: 63–68

[8] Oryan A, Zaker SR. Effects of topical application of honey on cutaneous wound healing in rabbits. Zentralbl Veterinaermed A 1998; 45: 181–188

[9] Ghelardini C, Galeotti N, Salvatore G, Mazzanti G. Local anaesthetic activity of the essential oil of Lavandula angustifolia. Planta Med 1999; 65: 700–703

[10] Hajhashemi V, Ghannadi A, Sharif B. Anti-inflammatory and analgesic properties of the leaf extracts and essential oil of Lavandula angustifolia Mill. J Ethnopharmacol 2003; 89: 67–71

[11] Cavanagh HMA, Wilkinson JM. Biological activities of lavender essential oil. Phytother Res 2002; 16: 301–308

[12] Lusby PE, Coombes AL, Wilkinson JM. A comparison of wound healing following treatment with Lavandula x allardii honey or essential oil. Phytother Res 2006; 20: 755–757

[13] Nada A, Abu-Ahmed H, Khafaga A, El-Kammar M. Clinical and histopathological evaluation of the effectiveness of lavender oil compared with black seed oil, ostrich oil and cod liver oil on the second intention wound healing in dogs. AJVS 2015; 46: 57

[14] Sipponen A, Jokinen JJ, Sipponen P, Papp A, Sarna S, Lohi J. Beneficial effect of resin salve in treatment of severe pressure ulcers: a prospective, randomized and controlled multicentre trial. Br J Dermatol 2008; 158: 1055–1062

[15] Sipponen A, Kuokkanen O, Tiihonen R, Kauppinen H, Jokinen JJ. Natural coniferous resin salve used to treat complicated surgical wounds: pilot clinical trial on healing and costs. Int J Dermatol 2012; 51: 726–732

[16] Jokinen JJ, Sipponen A. Refined spruce resin to treat chronic wounds: Rebirth of an old folkloristic therapy. Adv Wound Care 2016; 5: 198–207

[17] Sipponen A, Rautio M, Jokinen JJ, Laakso T, Saranpaa P, Lohi J. Resin-salve from Norway spruce – A potential method to treat infected chronic skin ulcers? DML 2007; 1: 143–145

[18] Rautio M, Sipponen A, Peltola R, Lohi J, Jokinen JJ, Papp A et al. Antibacterial effects of home-made resin salve from Norway spruce (Picea abies). APMIS 2007; 115: 335–340

[19] Sipponen A, Peltola R, Jokinen JJ, Laitinen K, Lohi J, Rautio M et al. Effects of Norway spruce (Picea abies) resin on cell wall and cell membrane of Staphylococcus aureus. Ultrastruct Pathol 2009; 33: 128–135

[20] Rautio M, Sipponen A, Lohi J, Lounatmaa K, Koukila-Kähkölä P, Laitinen K. *In vitro* fungistatic effects of natural coniferous resin from Norway spruce (Picea abies). Eur J Clin Microbiol Infect Dis 2012; 31: 1783–1789

[21] Sipponen P, Sipponen A, Lohi J, Soini M, Tapanainen R, Jokinen JJ. Natural coniferous resin lacquer in treatment of toenail onychomycosis: An observational study. Mycoses 2013; 56: 289–296

[22] Auvinen T, Tiihonen R, Soini M, Wangel M, Sipponen A, Jokinen JJ. Efficacy of topical resin lacquer, amorolfine and oral terbinafine for treating toenail onychomycosis. A prospective, randomized, controlled, investigator-blinded, parallel-group clinical trial. Br J Dermatol 2015; 173: 940–948

[23] Holmbom T, Reunanen M, Fardim P. Composition of callus resin of Norway spruce, Scots pine, European larch and douglas fir. Holzforschung 2008; 62: 444

[24] Goels T, Eichenauer E, Langeder J, Hoeller F, Sykora C, Tahir A et al. Norway spruce balm. Phytochemical composition and ability to enhance re-epithelialization *in vitro*. Planta Med 2020; 86: 1080–1088

[25] Heegaard PM, Stockmarr A, Piñeiro M, Carpintero R, Lampreave F, Campbell FM, Eckersall PD, Toussaint MJ, Gruys E, Sorensen NS. Optimal combinations of acute phase proteins for detecting infectious disease in pigs. Vet Res 2011; 42: 50

[26] Hennig-Pauka I, Menzel A, Boehme TR, Schierbaum H, Ganter M, Schulz J. Haptoglobin and C-reactive protein-non-specific markers for nursery conditions in swine. Front Vet Sci 2019; 6: 92

[27] Sipponen A, Laitinen K. Antimicrobial properties of natural coniferous rosin in the European pharmacopoeia challenge test. APMIS 2011; 119: 720–724

[28] Sipponen A, Jokinen JJ, Lohi J. Resin salve from the Norwegian spruce tree: A 'novel' method for the treatment of chronic wounds. J Wound Care 2007; 16: 72–74

[29] Spergser J, Loncaric I, Tichy A, Fritz J, Scope A. The cultivable autochthonous microbiota of the critically endangered northern bald ibis (Geronticus eremita). PLoS One 2018; 13: e0195255

[30] Lane DJ. 16S/23S rRNA Sequencing. In: Stackebrandt E, Goodfellow M, eds. Nucleic Acid Techniques in bacterial Systematics. Chichester: Wiley; 1991: 115–175

[31] Yoon SH, Ha SM, Kwon S, Lim J, Kim Y, Seo H, Chun J. Introducing EzBioCloud. A taxonomically united database of 16S rRNA gene sequences and whole-genome assemblies. Int J Syst Evol Microbiol 2017; 67: 1613–1617

[32] Stackebrandt E, Ebers J. Taxonomic parameters revisited. Tarnished gold standards. Microbiol Today 2006; 8: 6–9

[33] Stackebrandt E, Goebel BM. Taxonomic note: a place for DNA-DNA reassociation and 16S rRNA sequence analysis in the present species definition in bacteriology. Int J Syst Evol Microbiol 1994; 44: 846–849